

CineIQ Report

Purshotam Kumar

CineIQ: Hybrid Movie Recommendation Engine

Author: Purshotam Kumar

Date: May 2026

Code: github.com/proust-19/CINEIQ

1. Abstract

Content discovery on streaming platforms is opaque, biased toward promoted titles, and traps users in recommendation loops. CineIQ addresses this with an open, explainable hybrid movie recommendation engine combining content-based filtering (TF-IDF + cosine similarity), collaborative filtering (SVD matrix factorization), and sentiment-aware re-ranking – served via a FastAPI REST API and an interactive Streamlit dashboard.

2. Dataset

MovieLens 1M (grouplens.org/datasets/movielens/1m/): 1,000,209 ratings from 6,040 users on 3,883 movies across 18 genres, rated 1-5. Sentiment is proxied from rating values since review text is not available.

3. System Architecture

Content-Based Filtering: Genre labels are TF-IDF vectorized (English stop-word removal), producing a 3883 x 3883 cosine similarity matrix. Given a seed movie, the top-50 most similar candidates are retrieved.

Collaborative Filtering (SVD): An SVD model with 100 latent factors is trained for 20 epochs on the user-item matrix using the Surprise library. 3-fold cross-validation yields an RMSE of **0.8861**.

Weighted Ensemble: Scores from SVD (50%), Content (30%), and Popularity (20%) are min-max normalized and combined via weighted sum.

Sentiment Re-Ranking: Since ML-1M lacks review text, a proxy is derived (rating $\geq 4 = +1$, $\leq 2 = -1$, $3 = 0$), aggregated per movie with $\log(\text{review_count})$ confidence weighting, normalized to $[0,1]$. Final score = $0.7 \times \text{ensemble} + 0.3 \times \text{sentiment}$.

API & Dashboard: FastAPI serves `/recommend` and `/similar` endpoints. Streamlit dashboard visualizes results with Plotly bar charts and genre distribution pie charts.

4. Implementation Pipeline

Step	Notebook	Output
EDA	01_eda.ipynb	Statistics, visualizations, merged CSV
Content Model	02_content_based.ipynb	cosine_sim.pkl, indices.pkl

Step	Notebook	Output
SVD Model	03_collaborative_svd.ipynb	svd_model.pkl, RMSE = 0.8861
Ensemble	04_ensemble.ipynb	ensemble_weights.json
Sentiment	05_sentiment.ipynb	sentiment.pkl, movie_sentiment.csv

Notebooks run sequentially; each saves artifacts consumed by the next stage and the API.

5. Results

SVD 3-fold CV: Mean RMSE = 0.8861 (folds: 0.8862, 0.8859, 0.8861).

Qualitative (user=1, seed=“Toy Story (1995)”): Top results include Toy Story 2 (0.936), Wrong Trousers (0.922), Bug’s Life (0.918), Chicken Run (0.912), Iron Giant (0.897). Sentiment re-ranking further adjusts rankings by audience reception.

6. Future Work / Upgradation

- **Scale to MovieLens 25M** – retrain all models on 25x larger data for improved niche coverage
- **Deep Learning** – replace SVD with Neural Collaborative Filtering or two-tower models
- **Review Text Sentiment** – integrate actual IMDB reviews with DistilBERT/RoBERTa instead of rating proxies
- **Online Learning** – incremental FTRL or streaming SVD for real-time updates
- **Cold-Start** – use director, cast, plot embeddings (Sentence-BERT) for richer content features
- **A/B Testing** – experimentation layer to compare ensemble variants on engagement metrics

7. Conclusion

CineIQ demonstrates a modular, production-ready hybrid recommendation system achieving SVD RMSE of 0.8861 on MovieLens 1M. The weighted ensemble balances collaborative, content, and popularity signals, while sentiment re-ranking incorporates audience reception. The API-first design enables easy integration, and the proposed upgradation roadmap provides clear pathways for scaling and enhancement.

Report generated for CINEIQ project submission – May 2026